

Dependence of the volatility of world financial markets on the lunar phases, affecting the psychoemotional behavior of their participants

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This article examines the level of dependence of changes in various phases of the moon and an increase in intra-day volatility with simultaneous sharp price fluctuations in non-trading periods of BRENT oil futures on global financial markets over the past 28 years. A direct dependence has been identified between the behavior of participants of the on-exchange trading and changes in lunar phases, which influence psychoemotional behavior, fatigue, changes in state of health and, as a result, an increase in the instrument volatility, which in turn increases the risks for the participants themselves. The immediate cases of the greatest surges of activity, increased tension and stress have been identified. Phase changes have been identified as the most critical ones, with quite significant daily price gaps registered, where the risks can be controlled only by reducing the frequency and volume of transactions in advance, especially in case of overnight rollovers (weekends and holidays). The most important range of lunar phases has been specified: the period from the full moon phase to the new moon phase, especially the first day of these phases change. The findings are visualized in graphs that display average price gaps and intra-day price fluctuations across different data segments. Recommendations are also provided to reduce or completely prevent negative consequences.

Key words: moon phase, suicide, mood, mental state, mistakes, finance, oil, volatility, full moon, new moon, lunar cycle.

The Moon influence on all living and nonliving things has long been proven. The interposition of celestial bodies has an impact at the molecular level. Financial markets represent a poorly explored area of activity and price changes are sometimes even called chaotic, as not implying obvious dependencies, and here studies of the lunar phases can help us to some extent.

According to the market laws, the pricing of any asset is influenced by all the on-exchange trading participants. The current price at any given time is determined by the balance of supply and demand. At any given time, random, unpredictable events can also occur, e.g. failures associated with the technical equipment fails or malfunctions of communication networks. All these factors are undoubtedly reflected by current prices, since the participants are not only people, but also algorithms, both those with hard-coded logic and those based on self-learning neural networks. In turn, the imbalance here inevitably causes surges in volatility, which also has a temporary impact on the liquidity of the traded instrument. Trading in financial markets is a hyper-competitive activity; accordingly, the purpose and relevance of this article is to determine the related dependencies and to possibly prevent fatal actions of not only ordinary, but also professional on-exchange trading participants.

The Moon is the natural and only satellite of the Earth, located at a distance of 384,000 km, which is approximately 30 times the diameter of the Earth. The diameter of the Moon is slightly larger than a quarter of the Earth's diameter - 3,476 km.

Moon phases are gradually and cyclically changing light conditions over the period of a synodic month (about 29.5306 of an average solar day). The entire lunar month is conventionally divided into four parts, characterized by the position of the Moon on the Sun-Earth-Moon line: the new moon - a condition when the Moon is not visible, the first quarter - half of the Moon is visible, the full moon - the entire Moon is visible and the last quarter - half of the Moon is visible.

The human body contains a significant amount of water. With aging, this figure decreases, but on average, water amounts for approximately 70% of the human body, so our body follows the basic physical laws.

A study of the tap water properties during the new and full moon using wavelet analysis showed the dependence of changes in its properties on the moon phases. The amplitude of water particles vibrations increases and the multi-fractality of water also increases (its structuring decreases) during the new moon [3].

Some people react to the change of lunar phases with increased subjective feeling of loneliness from the last quarter of the lunar phase to the new moon, while the strongest reaction is observed during the full moon [1].

The moon influence on the mental condition has been determined by other researchers: during the new moon, people become more aggressive, and during the last quarter, increased agitation and uncontrollable behavior are registered [2]. The results of the study showed that the older a person is, the greater the influence the lunar phases have on their body [4].

During the new moon phase, the pressure of fluids in the body decreases. Emotional drop can adversely affect people prone to depression. During the first and last quarter, various types of imbalances are especially likely to develop in the body. General stress often causes headaches and increased nervousness. The full moon period can be difficult and critical for those prone to over-excitement and whose psyche is unstable. Our minds are so active at this time that it is difficult to fall asleep, so the full moon can create additional difficulties for people suffering from insomnia [4].

The most relevant article [5] examines a test of the simplest system for buying a portfolio of global indices during a new moon and selling during a full moon. It is more preferable to study the days when the moon phase changes; if the phase change occurs on the weekend, it is reasonable to assume that this will affect the subsequent first working day. It is this attitude towards tests that can help in developing a trading system component and, as a result, reduce the factor of errors and related risks.

According to many scientifically proven works, the phases of the new moon and full moon of our natural satellite are of greatest interest in this context. Let's first conduct a preliminary comparative analysis to confirm these conclusions and continue our research in a narrower direction.

The provided calculations use data for the period from 16.06.1992 to 04.08.2020 for BRENT futures contracts. The reason to choose this item is that oil is an essential life-sustaining resource that has been fought over for many years. This instrument is more than any other susceptible to various external influences, which can occur in the form of terrorist attacks on oil rigs and oil storage facilities, manipulation of news, escalation of armed conflicts on significant transport routes, the influence of the moon on health condition and behavior of people, including top officials of states whose decisions influence the negotiations results, e.g. OPEC.

The graph shown in Figure 1 displays the average values of gaps and intra-day price fluctuations for the entire period by day of the week. Let's use it as a base. The importance of the first working day of the week is clear. One can note a proof of the well-known fact about the often increased probability of a gap on Monday, which, coupled with the intra-day change, results in the maximum spread of prices compared to other days. It's not for nothing that they say Monday is surely worse than Sunday. On Wednesdays, statistical data on oil reserves in the United States is released.

Figure 2 represents the data in terms of moon phases. What stands out is the slightly increased average intra-day change during the new moon phase. Awareness of the fact that almost the same movements occur during the new moon phase as during other phases does not give any advantage in the market.

Average gaps and intra-day volatility by day of the week

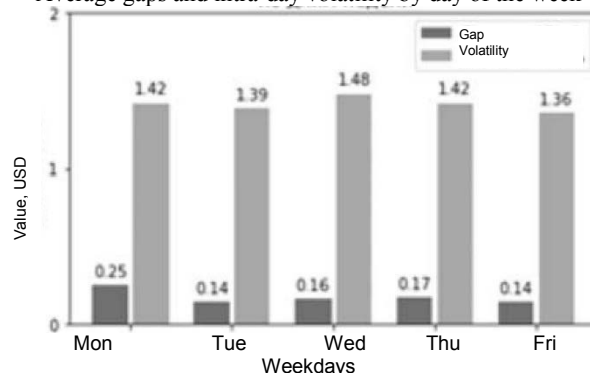


Figure 1. Average gaps and intra-day volatility by day of the week

Average gaps and intra-day volatility by moon phase

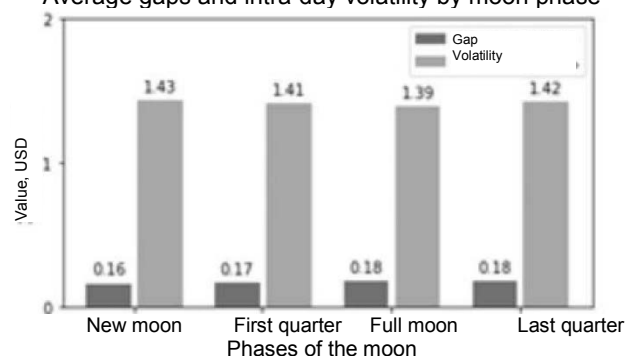


Figure 2. Average gaps and intra-day volatility by moon phase

Average gaps and intra-day volatility by day of the week, considering only phase changes

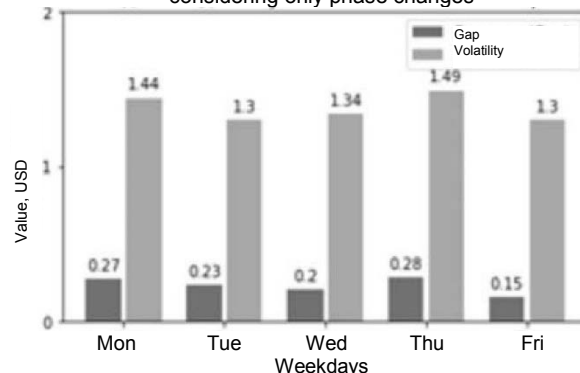


Figure 3. Average gaps and intra-day volatility by day of the week, considering only phase changes

Now let's take a look at the results, the calculations of which use only the first days during the changing phases of the moon. The graph shown in Figure 3 clearly shows the greater values for both indicators on Mondays and Thursdays. These findings confirm previous observations of gaps on the first working day after the weekend and the influence of changes in certain phases of the moon.

Figure 4 confirms the above description of the new moon and full moon phases features. The extreme values of the last quarter can also be considered, but one should consider the moment of transition to the new moon phase as the most significant one, further enhancing its effect. Thus, we note the time range most prone to increased volatility, starting from the full moon phase to transition to the new moon phase.

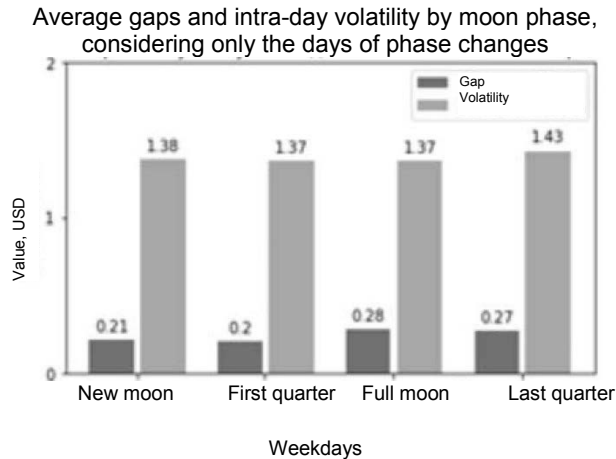


Figure 4. Average gaps and intra-day volatility by moon phase, considering only the days of phase changes

The example shown in Figure 5 already contains complete information about gaps by day of the week in the context of the moon phases, where one can find additional confirmation of the effectiveness of effect of the full moon, new moon and last quarter.

Figures 5 and 6 confirm previous research findings that volatility and gap risks are the highest on Monday, Thursday and Wednesday.

Another example (Figure 7) offers another proof of the importance of the first day of the moon phase change using the example of gaps and intra-day volatility (Figure 8).

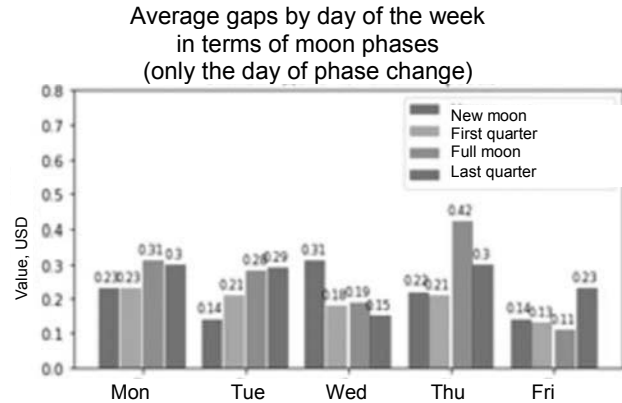


Figure 7. Average gaps by day of the week in terms of moon phases (only the day of phase change)

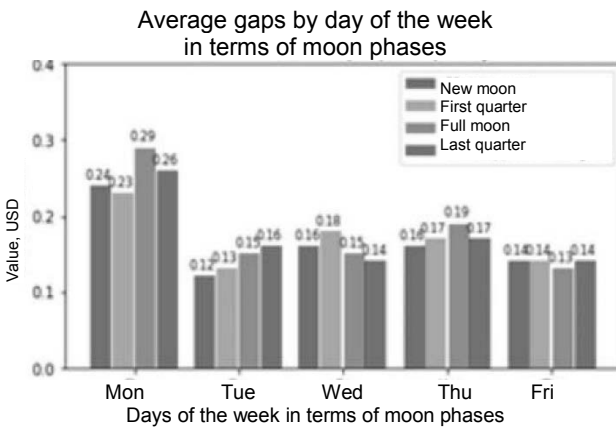


Figure 5. Average gaps by day of the week in terms of moon phases

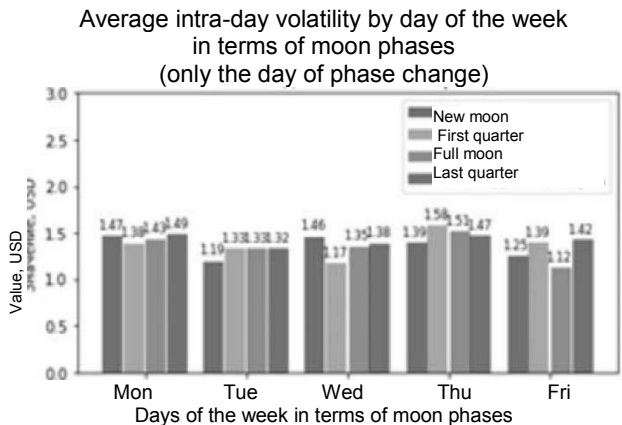


Figure 8. Average intra-day volatility by day of the week in terms of moon phases (only the day of phase change)

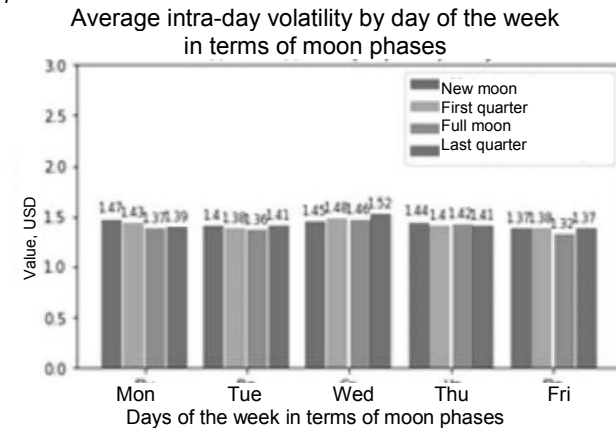


Figure 6. Average intra-day volatility by day of the week in terms of moon phases

Thus, based on the above research results, one can definitely conclude that various phases of the Earth's satellite have an impact on the behavior of the on-exchange trading participants and on all living things in general. The most volatile and therefore dangerous phases are the new moon and full moon, especially the days of transition to these phases, so one should be especially careful and stay as focused as possible to prevent various mistakes.

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